

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [112, 102, 96, 86, 85, 78, 69, 55, 61, 72]

Tap amplitudes (pixels): [144.62710672622887, 132.13629327327143, 130.59862173851607, 134.6291201783626, 132.90974381135493, 138.40520221436765, 128.96898852049665, 114.28473213863697, 88.40814442120137, 118.15244390193544, 122.88612614937456]

GPT-5

MDS-UPDRS Item 3.4 Scoring Analysis (Finger Tapping)

What the features suggest

- Taps: Reported “Number of taps: 11” with 10 periods listed. This is consistent with 11 peaks (amplitudes) over 10 intervals (periods) across the instructed 10 taps. I'll analyze the 10 intervals and the amplitude profile across the sequence.

Speed (periods in frames)

- Periods: [112, 102, 96, 86, 85, 78, 69, 55, 61, 72]
 - Clear speeding-up trend from the first to the 8th tap (112 → 55 frames), then slight variability (61, 72).
 - No evidence of slowing; if anything, there's a progressive speeding with minor end variability.
 - This pattern does not indicate mild/moderate slowing.

Rhythm/interruptions/hesitations

- Interruptions are typically seen as unusually long intervals or irregular halts compared to neighboring taps.
- Here, intervals decrease fairly smoothly, with small variations at the end (55 → 61 → 72). There are no conspicuous prolonged pauses or arrests. I do not see

3–5 interruptions, >5 interruptions, or a freeze. At most, there could be one minor hesitation in the last two intervals, but it's not clearly an interruption.

Amplitude and decrement

- Amplitudes: [144.6, 132.1, 130.6, 134.6, 132.9, 138.4, 128.97, 114.28, 88.41, 118.15, 122.89]
 - Early taps are large and stable (approx. 130–145) through about tap 6–7.
 - There is a notable decline starting around tap 7–8 with a low at tap 9 (≈ 88), then partial recovery at taps 10–11 (≈ 118 –123).
- Interpretation vs criteria:
 - “Amplitude decrements near the end” (score 1) vs “decrements midway” (score 2) vs “starting after the 1st tap” (score 3).
 - The amplitude is relatively preserved for the first 6–7 taps; the major drop occurs around taps 8–9, i.e., later in the sequence rather than midway. The recovery afterward also argues against a sustained, early decrement.
 - This pattern best matches “decrements near the end.”

Overall alignment with criteria

- No slowing; no clear multiple interruptions; no freeze.
- Amplitude decrement occurs near the end of the sequence.

Score: 1
